

# CONVOCATION QR REPORTING, ATTENDANCE & LIVE STAGE DISPLAY SYSTEM

## 48-Hour Development & Deployment Brief

*Prepared as a build-ready specification for the student development team*

|                    |                                                                                      |
|--------------------|--------------------------------------------------------------------------------------|
| Starting point     | Registered-student master database is already available                              |
| Primary goal       | One QR identity for reporting, attendance, stage verification and LED display        |
| Deadline           | Working MVP in Day 1; rehearsal-ready production system by end of Day 2              |
| Critical principle | Ceremony must continue even if QR, network or display system has a temporary failure |

**SUCCESS = Scan at reporting desk → mark present → pre-stage scan → operator-controlled LED display →  
final attendance/absence/conferred reports**

## 1. Scope: What Must Be Working After 48 Hours

- Import the existing registered-student database without re-entering student data.
- Generate one unique QR code/token for every registered student and provide a printable/downloadable QR pass.
- Scan the same QR at one or more Reporting Desks and mark the student as REPORTED with timestamp and desk/operator.
- Show a live dashboard: Registered, Reported, Yet to Report and reporting percentage, with School/Programme filters.
- Scan students at a Pre-Stage point, verify identity/sequence and place them into a display queue.
- Show the student's approved name, photograph, programme/degree, school and approved distinction/medal on the main LED screen.
- Keep public display changes under a Stage Operator's control; a scan must not blindly change the LED.
- Record stage completion/conferred status separately from reporting attendance.
- Export Registered, Reported, Absent, Conferred and Reported-but-Not-Conferred reports to CSV/XLSX.
- Provide a Help Desk search by Registration ID/PRN/name for damaged or unavailable QR codes.
- Maintain an audit log of important actions.

## 2. End-to-End Operational Flow

| Stage                    | Required behaviour                                                                                                                        |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A. Before event</b>   | Existing student master → validate required columns → generate opaque QR token → generate QR/pass → freeze approved display data.         |
| <b>B. Reporting</b>      | Student presents QR → desk scans → record appears with photo/name/programme → operator confirms → status becomes REPORTED with timestamp. |
| <b>C. Ceremony queue</b> | Only reported students are eligible for normal stage queue. Convocation sequence number determines order; QR only identifies the student. |
| <b>D. Pre-stage</b>      | Scan 3–5 students before stage → verify expected sequence → add student to display queue.                                                 |
| <b>E. Stage display</b>  | Stage Controller sees Current / Next / After Next → presses DISPLAY NEXT → audience LED changes to that student's slide.                  |
| <b>F. Completion</b>     | After degree presentation, mark CONFERRED/COMPLETED. If skipped, mark SKIPPED with reason.                                                |
| <b>G. Closure</b>        | Generate final attendance, absent, conferred and exception reports; preserve transaction log and database backup.                         |

## 3. Data Mapping From Existing Student Database

Do not redesign the registration database. Create an import/mapping layer. Minimum fields required:

| Existing field                  | Use in system                                             |
|---------------------------------|-----------------------------------------------------------|
| Registration ID / PRN           | Required; unique lookup key                               |
| Student Name                    | Required; approved convocation spelling                   |
| School / Institute              | Required                                                  |
| Department / Programme / Degree | Required as applicable                                    |
| Specialization                  | Optional                                                  |
| Photograph                      | Strongly recommended for reporting and stage verification |
| Award / Medal / Distinction     | Optional; only approved values displayed                  |
| Convocation Sequence No.        | Required for stage order                                  |
| Email / Mobile                  | Optional for pass distribution; never show publicly       |

System-added fields: QR Token, Registration Status, Reporting Status/Time/Desk, Late Flag, Pre-Stage Status/Time, Stage Status/Time, Degree Status, Remarks, Created/Updated timestamps.

## 4. Recommended 2-Day Technical Architecture

**Keep it simple.** Use one web application and one central database on a dedicated local network. Do not build the live ceremony around PowerPoint automation.

| Component   | Decision                                                                                          |
|-------------|---------------------------------------------------------------------------------------------------|
| Frontend    | React OR plain HTML/CSS/JavaScript. Use what the team already knows.                              |
| Backend     | FastAPI/Flask/Django OR Node/Express. Do not learn a new framework during these 48 hours.         |
| Database    | PostgreSQL/MySQL; SQLite acceptable only for a small single-server MVP after concurrency testing. |
| QR          | Opaque random token; QR contains token or application URL with token. No personal data inside QR. |
| Scanning    | USB QR scanner preferred; it behaves like keyboard input and is fast/reliable.                    |
| LED Display | Dedicated browser page in full-screen/kiosk mode connected by HDMI/LED controller.                |
| Networking  | Dedicated LAN/Wi-Fi. Prefer wired Ethernet for server and stage-control laptop.                   |
| Reports     | CSV is mandatory; XLSX desirable.                                                                 |

## 5. Core Database Logic

- student\_master: approved student data and convocation sequence.
- qr\_tokens: student\_id, opaque token, active flag, generated\_at.
- reporting: student\_id, reported\_at, desk\_id, operator\_id, late\_flag. Enforce one active reporting record per student.
- stage\_queue: student\_id, queued\_at, queue\_position/status.
- stage\_events: student\_id, displayed\_at, completed\_at, status (CONFERRED/SKIPPED), operator\_id.
- audit\_log: timestamp, user, student, action, device/desk and optional remarks.
- users: username, password hash, role, active status.

**Status rule:** REGISTERED → REPORTED → PRE-STAGE/QUEUED → DISPLAYED → CONFERRED. A registered student with no reporting record at closure is ABSENT. Never infer 'Conferred' merely from 'Reported'.

## 6. Screens the Team Must Build

| Screen              | Minimum requirement                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------|
| 1. Login            | Role-based login: Reporting, Help Desk, Stage Scanner, Stage Controller, Admin.                          |
| 2. Admin Import     | Upload CSV/XLSX; map columns; preview errors; import/update approved records.                            |
| 3. QR Generator     | Generate missing tokens; download QR sheet/pass; never regenerate existing token accidentally.           |
| 4. Reporting Desk   | Auto-focus scan box; show photo/name/programme/status; Confirm Reported; duplicate/invalid warnings.     |
| 5. Help Desk        | Search by PRN/Registration ID/name; manual verified reporting with mandatory reason/operator log.        |
| 6. Live Dashboard   | Counts and School/Programme filters; auto-refresh; export.                                               |
| 7. Pre-Stage        | Scan; show student + expected sequence; queue verified student; warn if not reported or out of sequence. |
| 8. Stage Controller | Current / Next / After Next; DISPLAY NEXT, HOLD, PREVIOUS, SEARCH, SKIP, COMPLETE.                       |

|                      |                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| <b>9. Public LED</b> | Clean 16:9 student slide only; no controls/private data; university holding screen between/when required. |
| <b>10. Reports</b>   | Registered, Reported, Absent, Conferred, Reported-not-Conferred, exceptions and audit export.             |

## 7. Reporting Desk Behaviour — Exact Rules

| Situation                       | System response                                                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Valid first scan                | Display student card → operator confirms → mark REPORTED → success sound/green confirmation → clear for next scan. |
| Duplicate scan                  | Show ALREADY REPORTED with original time/desk. Do not create duplicate attendance.                                 |
| Unknown QR                      | Show QR NOT RECOGNISED → direct to Help Desk.                                                                      |
| Inactive/cancelled registration | Show NOT ACTIVE → only authorized Help Desk/Admin can resolve.                                                     |
| No QR/damaged QR                | Help Desk searches student → verifies identity → manual reporting with reason logged.                              |
| Late arrival                    | Allow reporting but set LATE flag after configured cutoff.                                                         |

## 8. Stage Display Requirements

- Public screen format: 16:9, designed for readability from auditorium distance.
- Show only: university/convocation branding, approved student name, photograph, degree/programme, school and approved medal/distinction.
- Never show mobile number, email, internal database fields or unnecessary identifiers.
- Use a neutral university holding screen when no student is displayed.
- Preload/cache the next few student records/photos so the slide appears immediately.
- The public display must not change directly from a Reporting Desk scan.
- Stage Controller must have a one-key emergency HOLD/HOME action that returns the LED to the holding screen.

## 9. Required Reports and Definitions

| Report                     | Definition                                                                            |
|----------------------------|---------------------------------------------------------------------------------------|
| Registered                 | All active registered students in approved master.                                    |
| Reported / Present         | Active registered students with successful Reporting Desk transaction.                |
| Absent / Not Reported      | Active registered students without successful reporting transaction at final closure. |
| Late Reported              | Reported after configured cutoff.                                                     |
| Conferred                  | Students explicitly completed at stage.                                               |
| Reported but Not Conferred | Reported students without completed stage event.                                      |
| Exceptions                 | Invalid/duplicate/manual scans, skips and corrections.                                |

## 10. 48-Hour Execution Plan

| Timebox           | Deliverable                                                                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DAY 1 — 0–2 hrs   | Freeze scope. Inspect existing database. Confirm column mapping, student count, photos, sequence, medal fields. Assign team roles. Set up repository, server and DB. |
| DAY 1 — 2–6 hrs   | Build/import student master; generate QR tokens; build login and Reporting Desk; implement duplicate/invalid handling.                                               |
| DAY 1 — 6–10 hrs  | Build dashboard + attendance/absent queries + Help Desk search. Test simultaneous scans from multiple laptops.                                                       |
| DAY 1 — 10–12 hrs | Generate sample QR passes and freeze MVP. Backup code/database. End Day 1 only after reporting flow works end-                                                       |

|                          |                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                          | to-end.                                                                                                                             |
| <b>DAY 2 — 0–4 hrs</b>   | Build Pre-Stage queue, Stage Controller and 16:9 Public LED page.                                                                   |
| <b>DAY 2 — 4–7 hrs</b>   | Implement conferred/skipped events, reports, audit log, role permissions and holding screen.                                        |
| <b>DAY 2 — 7–9 hrs</b>   | Deploy on actual LAN/hardware; connect stage laptop to actual LED/projector; preload photos; configure backup laptop.               |
| <b>DAY 2 — 9–11 hrs</b>  | Full rehearsal with 30–50 dummy/student QRs including duplicate, invalid, absent, late, out-of-sequence and damaged-QR cases.       |
| <b>DAY 2 — 11–12 hrs</b> | Fix only critical issues; freeze build; take backup; print operator SOP; hand over credentials and final database/report procedure. |

## 11. Team Allocation for a 2-Day Build

| Role                            | Responsibility                                                              |
|---------------------------------|-----------------------------------------------------------------------------|
| <b>Team Lead / Integrator</b>   | Own scope, Git/repository, integration, deployment and final go/no-go.      |
| <b>Backend + Database (1–2)</b> | Import, API, statuses, reporting transactions, queue, reports, audit.       |
| <b>Frontend (1–2)</b>           | Reporting, Help Desk, dashboard, Stage Controller.                          |
| <b>Display/UI (1)</b>           | 16:9 LED template, holding screen, photo/layout optimization.               |
| <b>QR/Data (1)</b>              | QR token generation, pass output, data validation and sequence checks.      |
| <b>QA/Operations (1–2)</b>      | Test cases, multi-desk rehearsal, hardware/network checklist, operator SOP. |

## 12. Non-Negotiable Test Cases

- ☐ 1. Valid QR marks attendance once and records correct timestamp/desk/operator.
- ☐ 2. Same QR scanned twice does not create duplicate attendance.
- ☐ 3. Unknown QR is rejected cleanly.
- ☐ 4. Manual Help Desk attendance is logged with reason.
- ☐ 5. Two reporting desks can scan different students simultaneously without collision.
- ☐ 6. Dashboard totals reconcile: Registered = Reported + Not Reported (for active registrations).
- ☐ 7. Late scan is accepted and flagged correctly.
- ☐ 8. Unreported student scanned at Pre-Stage generates a warning and does not silently enter normal queue.
- ☐ 9. Out-of-sequence stage scan warns operator but permits authorized resolution.
- ☐ 10. LED displays correct student/photo/programme and no private information.
- ☐ 11. Stage completion creates Conferred record; reporting alone does not.
- ☐ 12. Reported-but-not-conferred report is accurate.
- ☐ 13. Server/network restart does not corrupt previous transactions.
- ☐ 14. Backup Stage Control laptop can take over.
- ☐ 15. Database backup can be restored and final reports regenerated.

## 13. Hardware / Venue Checklist

- Reporting laptops: number based on expected arrival rate; each with USB QR scanner.
- 1 Help Desk laptop + scanner; 1 Pre-Stage laptop + scanner.
- 1 Stage Control laptop connected to LED controller/projector; 1 configured backup laptop.
- Dedicated router/switch; wired Ethernet for server/stage where possible.
- Local server/main laptop with power adapter and UPS; network equipment on UPS.
- Spare scanner, HDMI cable/adapters, extension boards and printed student sequence list.
- One printed emergency manual attendance sheet and one printed stage sequence as last-resort fallback.

## 14. Security & Data Rules

- QR must use an opaque/random token; do not encode name, mobile, email or full student record.
- Role-based login; no shared administrator password at all counters.
- Password hashes, not plain-text passwords.
- Public LED endpoint receives only approved display fields.

- Record manual overrides and corrections in audit log.
- Take database backups before event, after reporting closes and after ceremony.
- Restrict exported student reports to authorized university staff and retain them per university policy.

## 15. Event-Day Operator SOP

### Reporting Operator

1. Login and confirm your desk number.
2. Keep cursor in Scan field; scan QR.
3. Verify photograph/name/programme; press Confirm Reported.
4. If Already Reported, do not override. If invalid or disputed, send to Help Desk.
5. Do not edit student master data at the reporting desk.

### Pre-Stage Operator

6. Scan student 3–5 positions before stage.
7. Confirm student and sequence warning, if any.
8. Add verified student to queue; never change approved name/degree from this screen.

### Stage Controller

9. Keep Current / Next / After Next visible on operator screen.
10. Use DISPLAY NEXT only after physical confirmation that the next student is approaching.
11. Use HOLD/HOME immediately if identity/order is uncertain.
12. After presentation, mark COMPLETE/CONFERRED; use SKIP only with reason.

### Administrator

13. Monitor totals and exceptions; do not make casual master-data changes during ceremony.
14. After cutoff, record late arrivals separately rather than deleting/reclassifying previous data.
15. At closure, export and reconcile all required reports and take final database backup.

## 16. Go/No-Go Acceptance Criteria

- ☐ 100% of active registered students are imported with unique IDs.
- ☐ 100% have unique QR tokens; sample printed/mobile QRs scan successfully.
- ☐ Reporting works from at least two simultaneous counters.
- ☐ Duplicate and invalid QR handling has been demonstrated.
- ☐ Dashboard and absent report reconcile with master counts.
- ☐ Pre-Stage → Stage Controller → LED works on the actual venue display chain.
- ☐ Correct student appears within approximately 1 second on local network under normal conditions.
- ☐ Stage operator can HOLD and recover from wrong/out-of-sequence scan.
- ☐ Final Conferred and Reported-but-Not-Conferred reports are demonstrated.
- ☐ Backup server/database and backup stage-control procedure are demonstrated.

## 17. Items to Freeze in the First 30 Minutes

**Project owner must give the team these decisions immediately; otherwise the 2-day deadline will slip:**

- Final registered-student database file and field mapping.
- Location of student photographs and naming convention.
- Final convocation sequence/order.
- Which fields appear on LED and exact university branding/background.
- Number of Reporting Desks and expected student count.
- Reporting cutoff time and definition of Late.
- Who is authorized to manually override/report a student.
- Venue network arrangement and LED/projector input specification.

- Final report formats required by university administration.

## 18. Explicitly Out of Scope for the 48-Hour Version

- New student registration/payment workflow — existing registered database is the source.
- Complex mobile app; use responsive browser screens.
- Facial recognition or biometric attendance.
- SMS/WhatsApp automation unless already available and trivial to integrate.
- Elaborate analytics beyond operational dashboard and required reports.
- Automatic PowerPoint generation/control; the live LED is a browser display.
- Cloud-only architecture that makes ceremony dependent on internet availability.

## 19. Final Handover Package

- Source code + README/deployment instructions.
- Database schema and clean backup.
- Environment/configuration file template without exposed production passwords.
- QR generation/pass procedure.
- Operator credentials/role matrix handed securely to authorized staff.
- One-page SOP for Reporting, Help Desk, Pre-Stage and Stage Controller.
- Test report showing all mandatory cases passed.
- Final executable/deployed build frozen after rehearsal.
- Emergency fallback procedure and printed sequence/attendance sheets.

## 20. Project Owner Sign-Off

|                                        |                          |
|----------------------------------------|--------------------------|
| Student master received                | <input type="checkbox"/> |
| LED design/data fields approved        | <input type="checkbox"/> |
| Reporting workflow approved            | <input type="checkbox"/> |
| Full rehearsal passed                  | <input type="checkbox"/> |
| Production build frozen / backup taken | <input type="checkbox"/> |

**The priority for this 48-hour project is reliability, speed and recoverability — not feature count.**